Conceptual Modeling

Library Management System — EECS447: Database Systems
Fall 2026 · Professor Hossein Saiedian
Team Name: DataBASED

1. Introduction

Project Overview:

> This database will be a reliable library management system that efficiently manages library books, magazines, digital media, client memberships, borrowing and returning transactions, reservations, and fees. The system will help library staff manage library operations and allow clients to search for items, reserve books, and check their loan status on their library membership. The database will also provide useful reports for both users for library fines, book availability, client activity, and more.

Scope:

> The library management system will manage books, digital media, and magazines; client accounts and membership types; borrowing and return transactions; reservations; and notifications for due dates, overdue items, and available reservations. Membership types will determine borrowing limits and fee rules. The system will generate reports on fines, item availability, borrowing trends, and client activity. It will include a staff interface for managing items, loans, returns, and accounts, and a client interface for searching the catalog, reserving items, and checking loan status. Work outside these library collection, circulation, and account functions is beyond the scope of this project.

2. ER Modeling Components

Entities:

- Item (Super class)
    - Book
    - Digital Media
    - Magazine
- User (Client)
- Membership

Relational Entities:

- Loan
- Reservation

Attributes:

- Item:
    - Item ID — **Primary key**
    - Title
    - Availability status
    - Restricted

- Book
    - Item ID — **Primary key, Foreign key from Item**
    - Author
    - Genre
    - Publication year
    - ISBN

- Digital Media
    - Item ID — **Primary key, Foreign key from Item**
    - Creator
    - Media type
    - Genre
    - Publication year

- Magazine
    - Item ID — **Primary key, Foreign key from Item**
    - Issue Number
    - Publication Date

- User (Client)
    - Client ID **— Primary Key**
    - First Name
    - Last Name
    - Contact Info
    - Account Status
    - Total Fees Owed
    - Membership Type — **Foreign key from Membership**

- Membership
    - Membership Type — **Primary Key**
    - Borrowing Limit
    - Daily Fine Rate
    - Loan Period
    - Reserve limit

- Loan (relational entity)
    - Loan ID — **Primary Key**
    - Client ID — **Foreign key from User**
    - Item ID — **Foreign key from Item**
    - Checkout Date
    - Due Date
    - Return Date
    - Fine Amount

- Reservation (relational entity)
    - Reservation ID — **Primary Key**
    - Client ID — **Foreign key from User**
    - Item ID — **Foreign key from Item**
    - Reservation Date
    - Reservation Status

Relationships:

ISA

- Item can be specialized as a book, digital media, or magazine

HAS

- Each user can only have one membership
- Each membership tier may have many clients
- A Loan makes reference to 1 item
- A Reservation makes reference to 1 item
- A Item can be mention between 0 to N number of reservations
- A Item can be mention between 0 to N number of loans

MAKES

- Users may borrow between 0 and `A` items
- Users may reserve between 0 and 15 items

NOTE: The value of `A` is based on the Membership attribute `Borrowing Limit`

Membership Types:

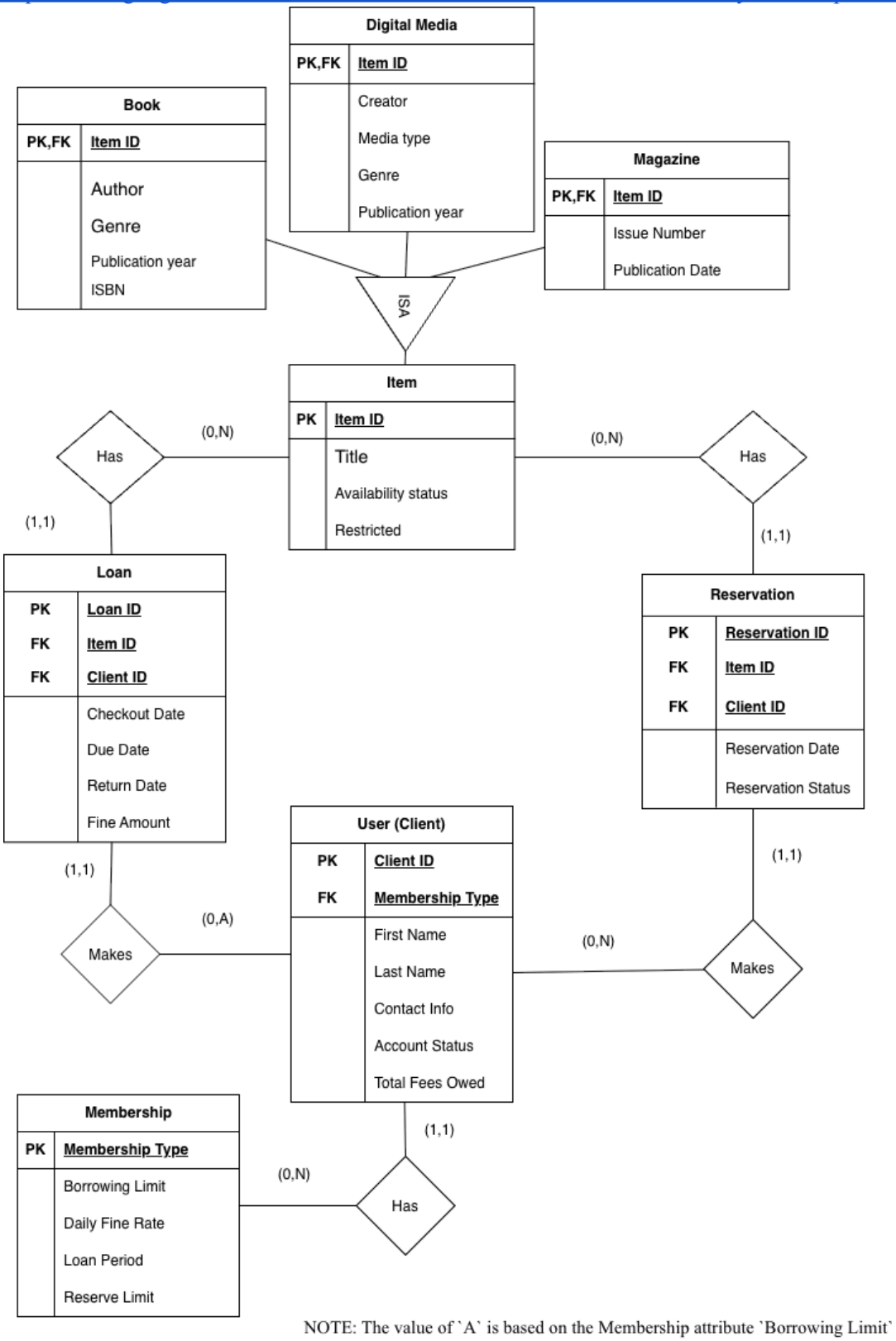
- Regular
    - Daily Fine Rate: $5
    - Borrowing Limit: 10 items
    - Loan Period: 2 weeks

- Student
    - Daily Fine Rate: $3
    - Borrowing Limit: 15 items
    - Loan Period: 3 weeks

- Senior
    - Daily Fine Rate: $3
    - Borrowing Limit: 10 items
    - Loan Period: 4 weeks

3. Create the ER Model

https://drive.google.com/drive/folders/1chZW5eF8LOi55M-m4x8tVznibE8yhCb1?usp=sharing



NOTE: The value of `A` is based on the Membership attribute `Borrowing Limit`